

Google Ads API — Tool Design & Architecture Documentation

Applicant Company: DataMaq (<https://datamaq.com.ar>) | **Developer Token:** Basic Access Application
| **Date:** August 2026

1. Company Overview & Business Model

DataMaq (<https://datamaq.com.ar>) is an industrial IoT, energy efficiency, and electrical telemetry engineering company based in Buenos Aires, Argentina. DataMaq installs hardware analyzers (Powermeter SmartPlus, IoT Gateways, and PLCs) in manufacturing facilities, helping PYMEs monitor energy consumption, eliminate reactive power penalties ($\cos \phi$), and avoid maximum demand surcharges.

Business Model: We operate under a field installation and engineering consulting model. Our cloud software platform is provided free (\$0 SaaS) to our clients. Revenue is generated exclusively from professional technical installation services, energy diagnostics, and ISO 50001 consulting.

2. Tool Description & Purpose

The tool being built is an **internal operational intelligence and budget-pacing monitor** (DataMaq Ads Auditor & MCP Tool). The tool connects exclusively to DataMaq's own Google Ads account to perform two core functions:

- **Automated Budget Pacing & Overrun Prevention:** Checks current daily ad spend against an internal limit (\$1,500 ARS/day) to ensure advertising campaigns stay within approved operational thresholds.
- **Search Terms & Keyword Quality Audit:** Analyzes real search queries from industrial users (e.g., 'multa factor de potencia', 'analizador de redes trifasico') to identify negative keywords and improve campaign quality scores.
- **Conversion Attribution Cross-Check:** Matches incoming B2B technical leads from our website contact forms with Google Ads click identifiers (GCLID) and UTM parameters.

3. Target Audience & User Access

User Access: Internal Users Only (DataMaq employees / technical operators).

The tool is strictly an internal administrative utility. It is not offered to third parties, clients, or the general public as a service. No external client accounts are managed or accessed through this developer token.

4. Architecture & Technical Design

Component	Technology / Implementation	Role / Function
Backend Core	Python 3.12+ / FastAPI / FastMCP	Executes internal queries and reporting tools via stdio / REST
Authentication	OAuth 2.0 (Offline Refresh Token)	Secure token-based auth with Google Ads API v18+
API Scope Used	https://www.googleapis.com/auth/adwords	Strictly read-only reporting queries (SELECT from campaign, customer, search_term_view)
Security Storage	Local encrypted environment (.env)	Secrets, client ID, and refresh token never exposed to clients

5. Data Flow & Privacy Compliance

1. The tool initiates a scheduled query using GoogleAdsClient with stored OAuth 2.0 refresh credentials.
2. Google Ads API processes the query via GoogleAdsService.SearchStream (GAQL).
3. Metrics (impressions, clicks, cost, conversions) are aggregated and checked against daily safety budgets.
4. Data is displayed exclusively to internal operators. **No data is stored on public databases or shared with third parties.**

6. Sample GAQL Queries Executed by the Tool

Budget Pacing Query:

```
SELECT metrics.cost_micros, metrics.clicks FROM customer WHERE segments.date DURING TODAY
```

Campaign Performance Query:

```
SELECT campaign.id, campaign.name, metrics.impressions, metrics.clicks, metrics.cost_micros, metrics.conversions FROM campaign WHERE segments.date DURING LAST_7_DAYS
```